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Identification of gallbladder cancer by direct near-infrared measurement of deuterated chloroform-extracted organic phase from human bile

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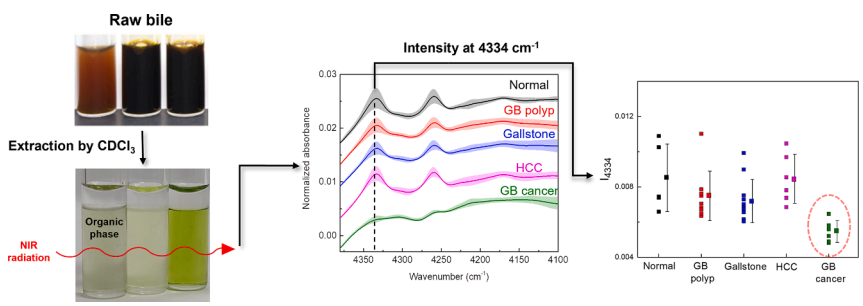
HIGHLIGHTS

- A direct NIR measurement of organic phase extracted from human bile was demonstrated.
- This scheme was to simply identify gallbladder (GB) cancer against other hepatobiliary diseases.
- $CDCl_3$ was used as an extraction medium for solvent interference-free NIR measurement.
- The peak intensities of the GB cancer samples were lower than those of the other diseases.
- The differences were statistically significant with a confidence interval greater than 99.0%.

GRAPHICAL ABSTRACT



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ABSTRACT

A simple near-infrared (NIR) spectroscopic scheme enabling direct measurement of organic phase extracted from human bile with no spectral interference from the extraction solvent was demonstrated for identification of gallbladder (GB) cancer. This scheme is used to recognize the different lipid contents in bile samples from GB cancer patients using NIR spectroscopy for disease identification. To this end, the extraction solvent should provide an absorption-free NIR region to observe peaks of related metabolite. For this purpose, deuterated chloroform ($CDCl_3$) is uniquely suited as an extraction medium because it has few absorption peaks in the 4380–4100 cm^{-1} range, where intense peaks for lipids and cholesterol are located. This exploratory study used 37 bile samples (obtained from five normal subjects and nine GB polyp, 11 gallstone, six hepatocellular carcinoma (HCC), and six GB cancer patients). The transmission NIR spectra of the organic phases extracted using $CDCl_3$ in a commercial glass vial were directly measured. The peak intensities of the GB cancer samples were lower than those of the other samples, and the differences were statistically significant, with a confidence interval greater than 99.0%. The lower lipid and cholesterol contents in the organic phases of the GB cancer samples were effectively identified in the corresponding NIR spectra. Therefore, the proposed NIR scheme is simpler and faster than the previous infrared (IR) measurement approach that requires solvent drying to highlight the buried metabolite peaks under a solvent absorption band.

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1. Introduction

Our research group explored vibrational spectroscopic methods for screening (identification) of gallbladder (GB) cancer using bile as a specimen [1,2]. As reported, GB cancer has a dismal prognosis, with a five-year survival rate of 6.4% [3,4]. Thus, early identification of GB cancer would be beneficial. The rationale behind our research was that bile composition can vary depending on the pathological condition of the patient. In our former report [1], the levels of lipids, such as phosphatidylcholine (PC) and phosphatidylethanolamine (PE), were lower in bile samples from the GB cancer group than in those from the normal/GB polyp/gallstone/hepatocellular carcinoma (HCC) groups based on the profiling results for nine lipid classes using the LC-MS. To recognize the low levels of PC and PE, the organic phase was extracted from bile using methanol: chloroform (CH_3OH:$CHCl_3$), and the transmission infrared (IR) spectra of the dried organic phases were analyzed to identify GB cancer.

Direct IR measurement of the extracted organic phase from bile was not possible due to complete extinction of IR radiation by the extraction solvent ($CHCl_3$). Therefore, the dried samples on an octadecyltrichlorosilane-coated Si wafer after evaporation of the solvent was unavoidably measured in a former study [1]. There were two weak points in the IR measurement of dried organic phase. First, the distribution of lipids and cholesterol in the dried spot was not homogeneous due to the so-called coffee-ring effect [5,6] occurring in solvent evaporation. Hence, multiple points on the dried sample had to be measured to ensure representative spectroscopic sampling. Second, delay of IR measurement was inevitable due to drying of the extracted organic phase. For these two reasons, the overall analysis was cumbersome and prolonged. Therefore, a simpler and faster spectroscopic scheme is needed.

The objective of this study was to develop a spectroscopic strategy for directly measuring the organic phase extracted from bile without drying. For this purpose, near-infrared (NIR) spectroscopy [7] was chosen as a potential alternative tool. Along with this study, NIR spectroscopy has been readily adopted for the identification of various diseases such as breast cancer[8], Alzheimer's disease[9], HIV infection [10], and diabetes[11], and diverse specimens (e.g., biopsy tissue and blood) were subsequently analyzed. Since NIR bands, which originate from overtone and combination vibrations, are considerably weaker in intensity and simpler in shape than IR bands, the existence of solvent absorption-free regions in the NIR range is possible. Therefore, deuterated chloroform ($CDCl_3$) was selected as the extraction solvent because it does not have absorption peaks in the 4380–4100 cm^{-1} range, where the major peaks of lipids and cholesterol are localized. Meanwhile, $CHCl_3$ has a strong absorption peak in this range and was not selected. In addition, to ensure simplicity of the measurement, an inexpensive disposable glass vial (inner diameter: 6.2 mm) was used for transmission NIR measurement in this study.

A total of 37 bile samples (obtained from five normal, nine GB polyp, 11 gallstone, six hepatocellular carcinoma (HCC), and six GB cancer subjects) was used. Each $CDCl_3$-extracted organic phase from those bile samples in a glass vial was subjected to NIR spectral acquisition. The transmission NIR spectral features, such as peak intensities, of the organic phases of the five groups were compared to determine the ability to discriminate the GB cancer group from the other groups. Next, the significance of the peak intensity differences among the five groups was statistically assessed using the Mann–Whitney *U* test. In addition, the NIR spectra of six lipid components (18:0–18:1 PE, 18:0–18:1 PC, 18:0 lysophosphatidylcholine (LysoPC), linoleic acid, palmitic acid, and cholesterol) were examined to elucidate the peaks in the organic-phase spectra. Finally, the discriminability achieved with the NIR measurement was compared with that in a previous IR measurement.

2. Experimental section

2.1. Collection of bile samples and extraction of their organic phases

A total of 37 bile samples was collected from five normal, nine GB polyp, 11 gallstone, six HCC, and six GB cancer patients after cholecystectomy at Hanyang University Hospital (Seoul, Republic of Korea). The purpose and content of the research were thoroughly explained to each subject. The study was performed under guidelines from the Institutional Review Board of Hanyang University (HYI-18–227-15). The bile samples were frozen immediately after collection and stored at less than − 20 °C to prevent further denaturation. Each bile sample was thawed at room temperature before extraction of the organic phase.

For organic-phase extraction, each bile sample (50 μL) was mixed with $CDCl_3$:CH_3OH:NaCl (0.73%) (10:3:3) solution (800 μL) in a microtube. The extraction mixture allowed simple lipid extraction, as previously reported [12]. The mixture of bile and extraction solution was centrifuged for 5 min at 2000 RPM. After centrifugation, the bottom organic phase containing fatty acids, cholesterol, and phospholipids was separated, and 250 μL of that organic phase was transferred into a cylindrical glass vial (inner diameter: 6.2 mm, outer diameter: 8.2 mm, obtained from Wheaton). The organic phase in the glass vial was further centrifuged with $CDCl_3$:NaCl (0.73%) (5:2) solution (350 μL) for 5 min at 2000 RPM and used for NIR spectral acquisition. The $CHCl_3$, $CDCl_3$, and CH_3OH were purchased from Sigma-Aldrich.

2.2. Acquisition and analysis of NIR spectra

All NIR spectra of the $CDCl_3$-extracted organic phases in glass vials were collected using an MB-160 FT-NIR spectrometer (ABB Analytical Measurements, Quebec, Canada) equipped with a glass vial holder. Because the surface of the glass vial was round, a narrow rectangular aperture (width: 0.85 mm) was positioned just in front of the glass vial to minimize the undesired distortion of spectral features, such as a nonlinear baseline. Each spectrum was collected by adding 64 scans with a resolution of 4 cm^{-1}. Baseline correction of the raw spectra and the Mann–Whitney *U* test were performed using MATLAB R2020a (MathWorks, Inc., Natick, MA, USA).

All pure components examined in this study were purchased from well-known suppliers (18:0–18:1 PE, 18:0–18:1 PC, 18:0 LysoPC: Avanti Polar Lipids, Inc.; cholesterol and linoleic acid (liquid): Sigma-Aldrich; palmitic acid: Tokyo Chemical Industry). To acquire NIR spectra for the pure solid components (18:0–18:1 PE, 18:0–18:1 PC, 18:0 LysoPC, palmitic acid, and cholesterol), a proper amount of each sample was mixed with KBr powder, and a pellet prepared from that mixture was analyzed. The liquid (linoleic acid) sample was transferred into a quartz cuvette with a pathlength of 0.5 mm for NIR measurement. The baseline of each pure component spectrum was corrected to zero at 5000 cm^{-1}, and the baseline-corrected spectrum was normalized to a maximum absorbance of 1.0 for the band with the highest absorbance.

3. Results and discussion

3.1. Spectral features of $CHCl_3$, $CDCl_3$, lipids, and cholesterol

In our previous report [1], $CHCl_3$ was used as the solvent to extract the organic phase containing hydrophobic metabolites (lipids and cholesterols) from bile for drop-and-dry IR measurement. It was beneficial to examine the locations of the NIR peaks of $CHCl_3$, especially in the 5000–4000 cm^{-1} range (the so-called combination range), which is the most sensitive NIR region and is located just before the mid-IR region. If $CHCl_3$ provides an absorption-free region where extracted metabolites can be recognized, direct NIR measurement of extracted organic phases without solvent drying is feasible. Fig. 1 (a) shows the spectrum of $CHCl_3$ in a glass vial in the 5000–4000 cm^{-1} range. As shown, a strong and broad peak appears around 4250 cm^{-1} and is saturated due to the thick

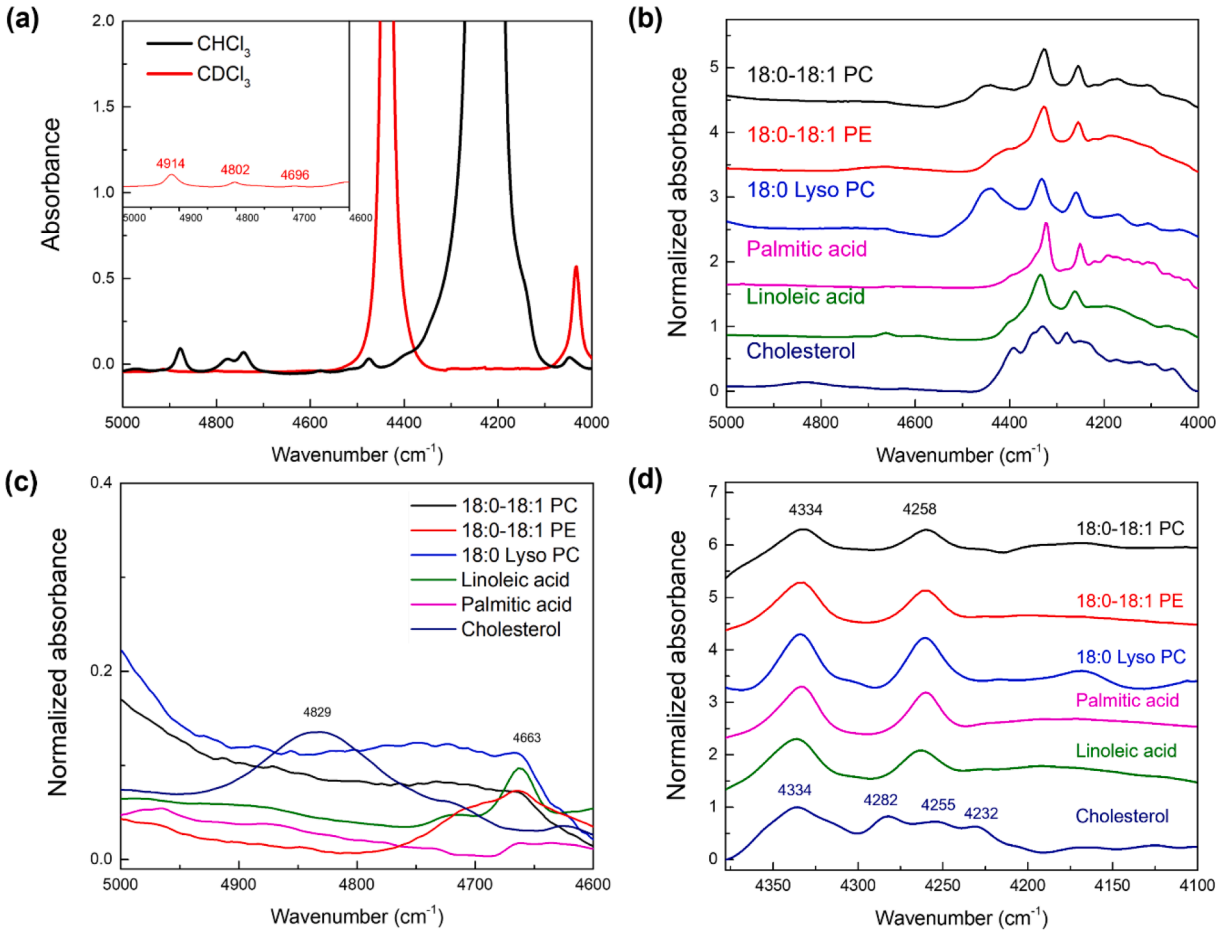

Fig. 1. NIR spectra of $CHCl_3$ and $CDCl_3$ (a) in the 5000–4000 cm^{-1} range. The inset highlights the $CDCl_3$ peaks in the 5000–4600 cm^{-1} range. NIR spectra of 18:0–18:1 PC, 18:0–18:1 PE, 18:0 LysoPC, palmitic acid, linoleic acid, and cholesterol in the 5000–4000 cm^{-1} (b) and 5000–4600 cm^{-1} range (c), and the same pure components separately dissolved in $CDCl_3$ (d).

pathlength (6.2 mm). As reported, $CHCl_3$ has four IR bands [13–16]: C–H stretching at 3019 cm^{-1}, C–H deformation at 1215 cm^{-1}, C-Cl asymmetric stretching at 769 cm^{-1}, and C-Cl symmetric stretching at 681 cm^{-1}. Therefore, the 4250 cm^{-1} peak originates from the combination of the two peaks at 3019 and 1215 cm^{-1} (3019 cm^{-1} + 1215 cm^{-1} = 4234 cm^{-1}, close to 4250 cm^{-1}). Furthermore, small peaks are located at 4877, 4742, 4474, and 4050 cm^{-1}. The intensities of the three peaks in the 5000–4400 cm^{-1} range are weak, so NIR spectroscopic recognition of peaks for extracted metabolites is not problematic in this range.

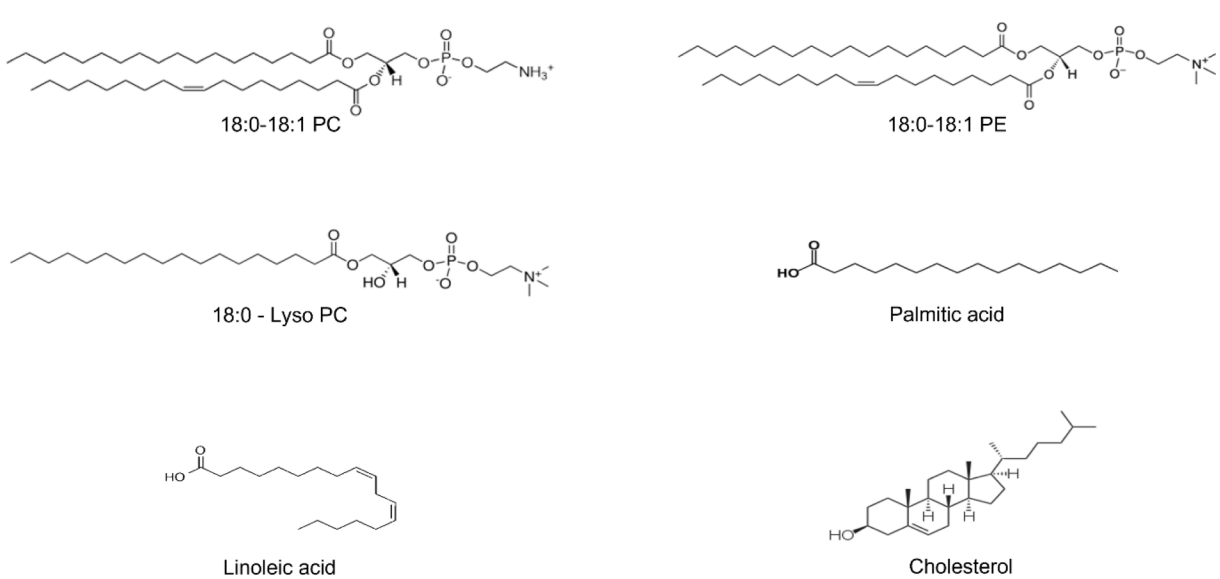

Fig. 2. Molecular structures of 18:0–18:1 PC, 18:0–18:1 PE, 18:0 LysoPC, palmitic acid, linoleic acid, and cholesterol.

Initially, NIR spectral features of major components extracted in the organic phase were acquired to examine where their bands were located in the 5000–4000 cm^{-1} range. Fig. 1 (b) displays the NIR spectra of five representative pure lipids (18:0–18:1 PC, 18:0–18:1 PE, 18:0 LysoPC, palmitic acid, and linoleic acid) and cholesterol. Each of the spectra is properly offset for comparison. The molecular structures of these components are shown in Fig. 2. The NIR peaks of each component appear across the whole range. The intense peaks are rather localized in the 4380–4100 cm^{-1} range, such as the two peaks around 4334 and 4258 cm^{-1} that originate from aliphatic hydrocarbon chains. The combination of CH_3 asymmetric stretching and CH_2 bending is responsible for the 4334 cm^{-1} peak, and the 4258 cm^{-1} peak originates from a combination of CH_3 symmetric stretching and CH_2 bending [17–20]. Meanwhile, the peak shapes of cholesterol are somewhat different from those of the other components due to its different molecular structure. Unfortunately, as the $CHCl_3$ peak appears in the 4380–4100 cm^{-1} range, NIR features of lipids and cholesterol cannot be recognized when using $CHCl_3$ for extraction. Therefore, extraction solvents with no or minimal NIR absorption in the 4380–4100 cm^{-1} range and an acceptable extraction ability are required.

A reasonable choice of solvent would be deuterated chloroform ($CDCl_3$) because it has nearly the same extraction ability as $CHCl_3$ and its peak would be located away from the 4380–4100 cm^{-1} range due to the varied molecular vibrations by the substituted deuterium. Fig. 1 (a) also shows the NIR spectrum of $CDCl_3$, comprising a strong peak at 4450 cm^{-1} and a weak peak at 4025 cm^{-1} but no peaks in the 4380–4100 cm^{-1} range. Therefore, the lipid and cholesterol peaks can be highlighted when $CDCl_3$ is used as the extraction solvent. As reported, $CDCl_3$ has four major peaks in the IR region [15,16,21,22]: C-D stretching at 2256 cm^{-1}, C-D bending at 914 cm^{-1}, C-Cl asymmetric stretching at 744 cm^{-1}, and C-Cl bending at 655 cm^{-1}. The 4450 cm^{-1} peak of $CDCl_3$ corresponds to the first overtone of C-D stretching at 2256 cm^{-1} (2256 $cm^{-1} \times 2 = 4512$ cm^{-1}, close to 4450 cm^{-1}). The peak at 4025 cm^{-1} originates from the combination of C-D stretching at 2256 cm^{-1} and the first overtone of C-D bending at 914 cm^{-1} (2256 $cm^{-1} + (2 \times 914$ $cm^{-1})$ $= 4084$ cm^{-1}, near 4025 cm^{-1}). In addition, the inset magnifies the minute peaks of $CDCl_3$ at 4914, 4802, and 4696 cm^{-1}.

Fig. 1 (c) presents the NIR spectra of the same pure components in the 5000–4600 cm^{-1} range, where the absorption of $CDCl_3$ is nearly negligible as shown in Fig. 1 (a). As seen, the peaks of pure components also appear in this range, while their magnitudes are much weaker and less characteristic than those in the 4380–4100 cm^{-1} range. Therefore, the spectral features in this range would be less informative for subsequent discriminant analysis.

3.2 NIR spectral features of lipids/cholesterol in $CDCl_3$ and the $CDCl_3$-extracted organic phases

Fig. 1 (d) shows the NIR spectra (4380–4100 cm^{-1} range) of 18:0–18:1 PC, 18:0–18:1 PE, 18:0 LysoPC, palmitic acid, linoleic acid, and cholesterol that were separately dissolved in $CDCl_3$ solution. In comparison with the spectra of the same components that are not dissolved in the $CDCl_3$ (Fig. 1 (b)), the bands are broader and have less detailed features due to molecular interactions with the surrounding $CDCl_3$ molecules. The 4334 and 4258 cm^{-1} peaks are again apparent for the five lipids, and their band shapes and positions differ slightly from one another. The peak shapes of cholesterol are again dissimilar from those of the lipid components. This observation confirms that $CDCl_3$ absorption does not seriously obscure the peaks of the dissolved lipids and cholesterol.

Fig. 3 (a) shows the average NIR spectra of the organic phases extracted from the bile samples of the normal (black), GB polyp (red), gallstone (blue), HCC (magenta), and GB cancer (green) groups. Each average spectrum is arbitrarily offset for comparison. Before averaging, the baseline of each raw spectrum was vertically corrected to zero at 4380 cm^{-1}. The shades alongside each average spectrum represent the standard deviations of absorbance at each wavenumber, indicating the spectral variation among the samples in each group. Similar to the spectra of the pure components (Fig. 1 (d)), 4334 cm^{-1} and 4258 cm^{-1} peaks were observed in the average spectra of all five groups. The intensities of the two peaks in the normal, GB polyp, gallstone, and HCC groups are analogous to each other and larger than those in the GB cancer group. As mentioned in the Introduction, the lipid contents are low in the bile samples from GB cancer subjects due to dilution of bile

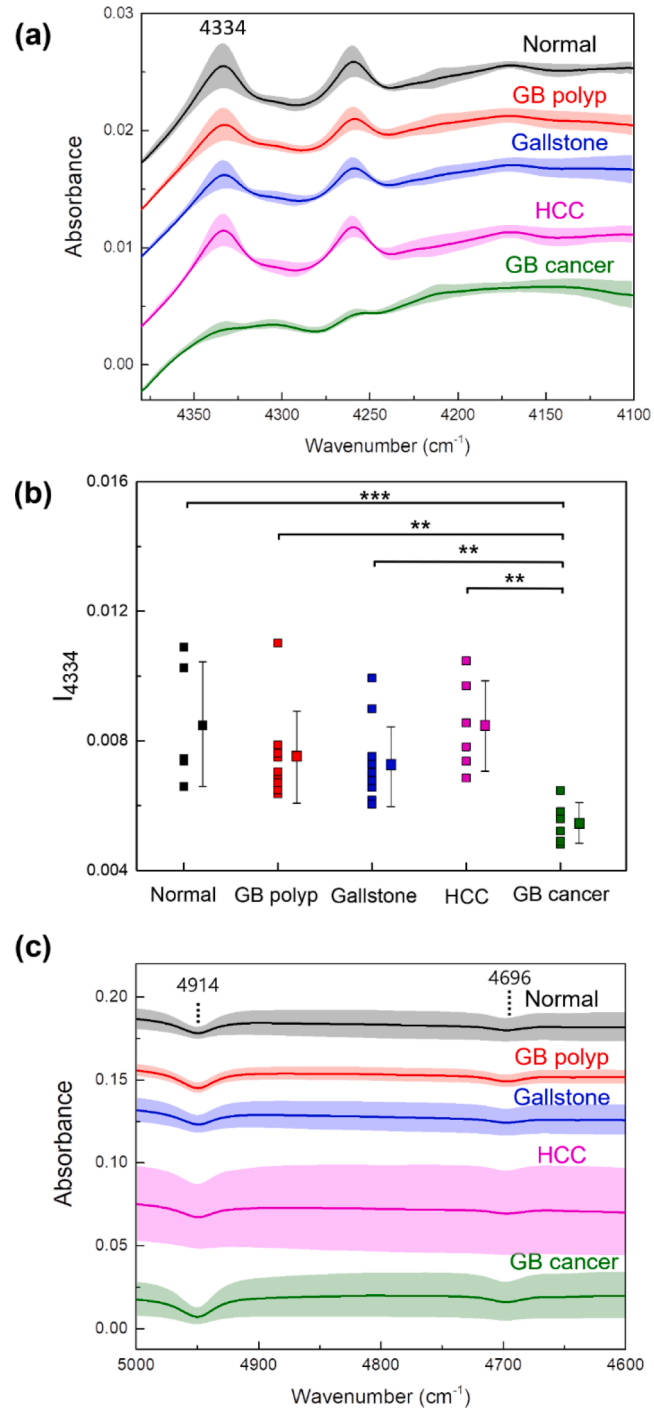

Fig. 3. Average NIR spectra of the organic phase extracted from bile samples of the normal, GB polyp, gallstone, HCC, and GB cancer groups in the 4380–4100 cm^{-1} (a) and 5000–4600 cm^{-1} range (c). The shades alongside each average spectrum represent the standard deviations of absorbance at each wavenumber, indicating the spectral variation among the samples in each group. The I_{4334} values in the sample spectra of each group is also shown (b).

caused by the decreased water-absorption ability of the thickened GB wall [23,24]. Therefore, the decreased concentrations of lipids and cholesterol in the organic phases of GB cancer samples are successfully reflected in the corresponding NIR spectra.

For quantitative evaluation, the intensity of the 4334 cm^{-1} peak (I_{4334}) in the sample spectra of each group were obtained, and the average and standard deviation (vertical bar) of those I_{4334} values were analyzed, as shown in Fig. 3 (b). The difference in I_{4334} values between the pairs of groups was statistically evaluated using the Mann–Whitney *U* test, which was executed for all 10 combinatorial cases of two-group comparison. The pairs of groups showing significant differences are designated by ** or ***, corresponding to *p*-values smaller than 0.01 (99.0% confidence interval (CI)) or 0.001 (99.9% CI), respectively. The I_{4334} values for the GB cancer samples are lower than those for the normal, gallstone, GB polyp, and HCC samples, with CIs greater than 99.0%. On the other hand, the I_{4334} values do not differ among the remaining four groups (no designated asterisks). Overall, differentiating GB cancer samples by comparing the I_{4334} value against those of samples from the other four groups is feasible with a 99.0% CI.

The average NIR spectra of the five sample groups in the 5000–4600 cm^{-1} range were additionally examined as shown Fig. 3 (c). Each average spectrum is arbitrarily offset for comparison. The absorption peaks corresponding to the extracted components are hardly observable in this range, while the minute spectral variations are occurred around 4914 and 4696 cm^{-1}, corresponding to the locations of $CDCl_3$ absorption peaks (refer to the inset in Fig. 1 (a)). It informs that the $CDCl_3$-induced spectral variations are only featured in this range rather than the absorptions of extracted components themselves. Therefore, the spectral features in this range are not useful to discriminate the GB cancer group against the remained groups.

3.3. Comparison of discriminability between NIR and IR measurements

It was interesting to compare the discriminability of GB cancer based on NIR measurement of the $CDCl_3$-extracted bile samples with that in the previous IR analysis using dried organic phases extracted from bile samples. Meanwhile, since the current and previous studies were executed in different periods, the number of co-employed bile samples in both studies was 26 (five normal, seven GB polyp, five gallstone, five HCC, and four GB cancer samples), approximately 70% of the total samples. Some of the samples were fully consumed in the former study and new samples were added for the current study. Therefore, the comparison was performed using the 26 co-employed samples.

Fig. 4 (a) shows the average normalized IR spectra of dried organic phases extracted from the co-employed samples belonging to each group. Before averaging the spectra in each group, the baseline of each raw spectrum was corrected to zero at 1800 and 600 cm^{-1}, and the baseline-corrected spectrum was divided by the corresponding peak area in the 1800–600 cm^{-1} range for normalization. The peak at 1738 cm^{-1} corresponds to the C = O group in phospholipids, and the 1465 and 1377 cm^{-1} peaks are from the CH_2 bending and symmetric stretching vibration of cholesterols as well as various lipids [25–27]. As observed, the intensities of the 1465 and 1377 cm^{-1} peaks are relatively higher for the GB cancer samples due to the area normalization of the spectra.

Fig. 4 (b) shows the I_{1465} values in the IR spectra of the co-employed samples in each group. The I_{1465} values of the GB cancer samples are higher than those of the other samples. The differences in I_{1465} values between the GB cancer group and each of the remaining groups are all significant with a 95% CI (*p*-values smaller than 0.05). Fig. 4 (c) presents the I_{4334} values in the NIR spectra of the same co-employed samples. The differences in the I_{4334} values between the GB cancer group and each of

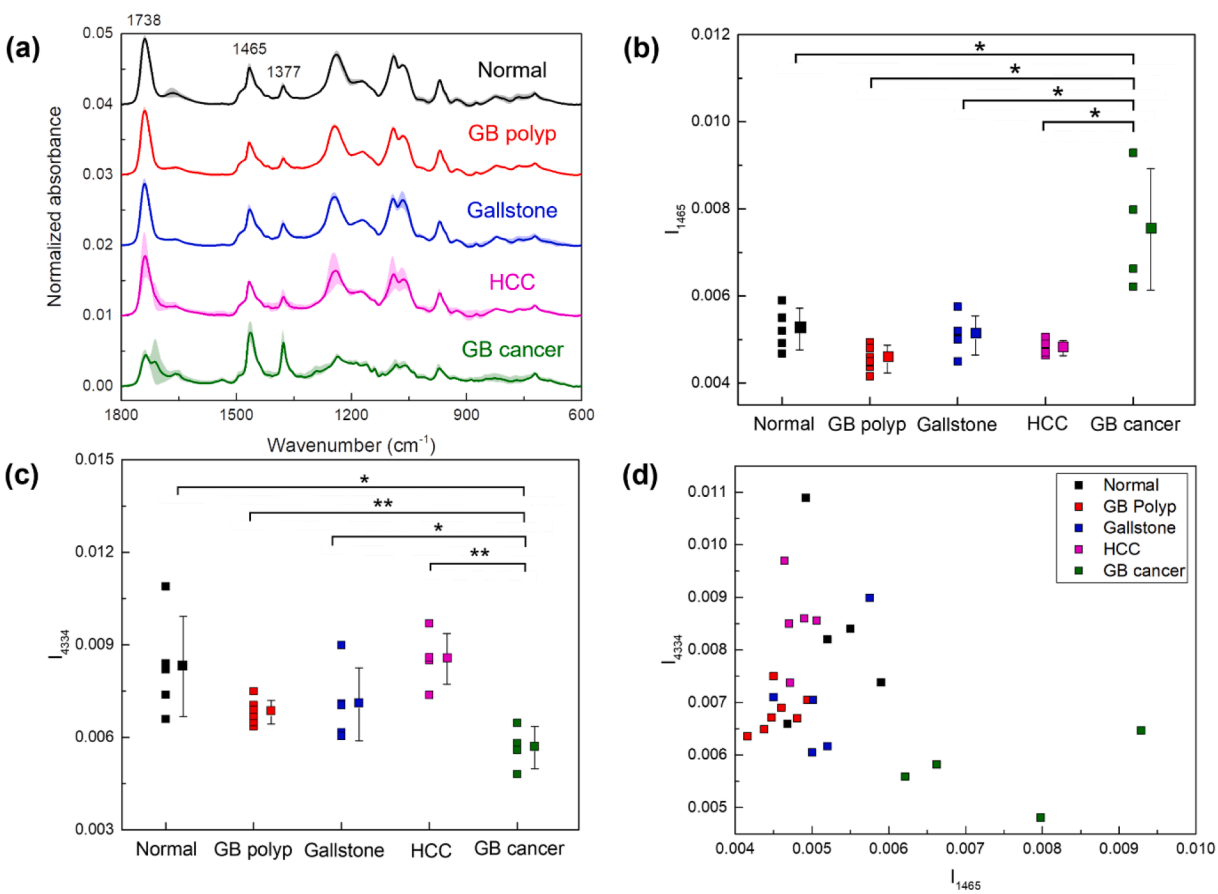

Fig. 4. Average IR spectra of the dried organic phases extracted from the 26 co-employed bile samples in each group (a) and the corresponding I_{1465} values (b). The I_{4334} values in the NIR spectra of the same samples (c) and the two-dimensional distribution of I_{1465} vs. I_{4334} values (d).

the remaining groups are all significant with either a 95% or 99.0% CI. In the two-dimensional domain of I_{1465} vs. I_{4334} values (Fig. 4 (d)), the four GB cancer samples are located at the right bottom and apart from the samples of the other groups. This observation confirms that the different contents of lipids and cholesterols in the organic phase extracted from the bile samples of GB cancer were clearly identified by both IR and NIR measurements. Meanwhile, the NIR measurement was simpler and faster by direct analysis of the $CDCl_3$-extracted bile sample without the solvent-drying step, which was unavoidable in the IR measurement.

4. Conclusion

The low lipid and cholesterol contents in bile samples from GB subjects were effectively identified by NIR spectroscopy using unique $CDCl_3$-based extraction of raw bile. The NIR bands of the extracted metabolites were clearly observed in the 4380–4100 cm^{-1} range, corresponding to the transparent NIR window provided by $CDCl_3$. The I_{4334} values of the GB cancer samples were lower than those of the normal, GB polyp, gallstone, and HCC samples, and the differences were statistically significant. Therefore, identifying GB cancer by analyzing bile samples using the demonstrated spectroscopic scheme is feasible. Furthermore, the presented NIR scheme facilitates simpler and faster analysis of bile samples compared to the previous IR scheme based on drop-and-dry measurement.

In addition to feasibility, two important issues must be further addressed for clinical applicability of the proposed scheme. First, a sample set with a larger number of bile samples is required for additional clinical assessment. However, collection of bile samples from GB cancer patients is time-intensive because of the low incidence rate. Second, a one-pot extraction method that is executable in a glass vial would be versatile, further shorten the analysis time, and reduce labor, allowing high-throughput analysis. Sample collection is currently underway, and new extraction methods are under development.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

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